MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG4DSECSC202 - SYSTEM PROGRAMMING

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
(Answer all questions, each question carries 2 marks.)

1. Define system software. [U] [CO1]
2. Indicate the use of debug monitors in program development. [U] [CO2]
3. Distinguish between linking and loading. [U] [CO3]
4. Indicate where code optimization is applied in compilation. [U] [CO2]
5. State the purpose of overlays in linking. [U] [CO3]

[5 x 2 = 10]

Part B
Short Answer Questions
(Answer any 5 questions, each question carries 5 marks.)

6. Explain the major functions of system software with suitable examples. [U] [CO1]
7. Explain the process of parsing and discuss its role in language processing. [U] [CO1]
8. Demonstrate macro definition and macro call using a suitable example. [A] [CO2]
9. Illustrate how control structures are translated during compilation. [A] [CO2]
10. Analyze the working of different types of loaders. [An] [CO3]
11. Examine the role of a linker in resolving external references. [An] [CO3]
12. Explain the design and working of a Two-Pass Assembler. [U] [CO1]

[5 x 5 = 25]

Part C
Essay Questions
(Answer any 1 question, each question carries 15 marks.)

13. Apply macro processing concepts to design a Macro Preprocessor, explaining parameter handling and expansion mechanisms. [A] [CO2]

14. Analyze the design of Linkers and Loaders, and evaluate their role in relocation, linking, and program execution. [An] [CO3]

[1 x 15 = 15]